

Compiler Design Lab Programs

Name: Renee Vincent A.S.R

Reg No: 192424161

Course Code: CSA1402

23. Write a LEX program to identify the capital words from the given input.

Code:

```
%{  
#include <stdio.h>  
%}  
%%  
[A-Z]+ { printf("Capital Word: %s\n", yytext); }  
.\n ;  
%%  
  
int main(){ yylex(); return 0; }  
  
int yywrap(){ return 1; }
```

Output:

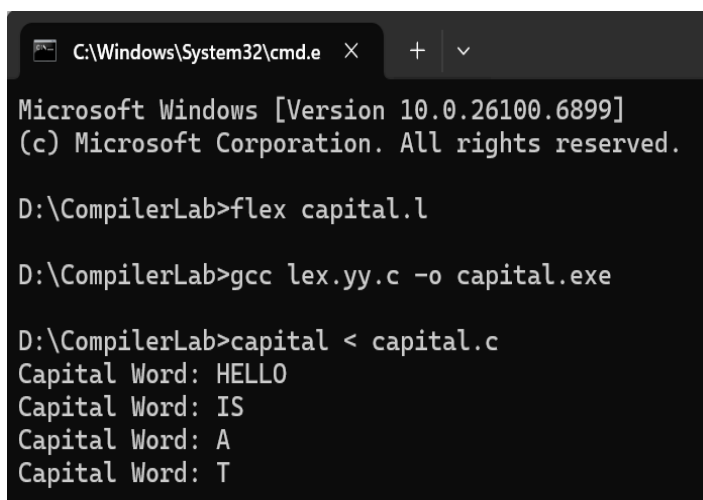
HELLO this IS A Test

Capital Word: HELLO

Capital Word: IS

Capital Word: A

Output Screenshot:



```
C:\Windows\System32\cmd.e × + v  
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\CompilerLab>flex capital.l  
  
D:\CompilerLab>gcc lex.yy.c -o capital.exe  
  
D:\CompilerLab>capital < capital.c  
Capital Word: HELLO  
Capital Word: IS  
Capital Word: A  
Capital Word: T
```

24. Write a LEX Program to check the email address is valid or not.

Code:

```
%{  
#include <stdio.h>  
%}  
%%  
[a-zA-Z0-9._%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,} { printf("Valid Email: %s\n", yytext); }  
[^\n]+ { printf("Invalid Email: %s\n", yytext); }  
\n ;  
%%  
  
int main(){ yylex(); return 0; }  
int yywrap(){ return 1; }
```

Output:

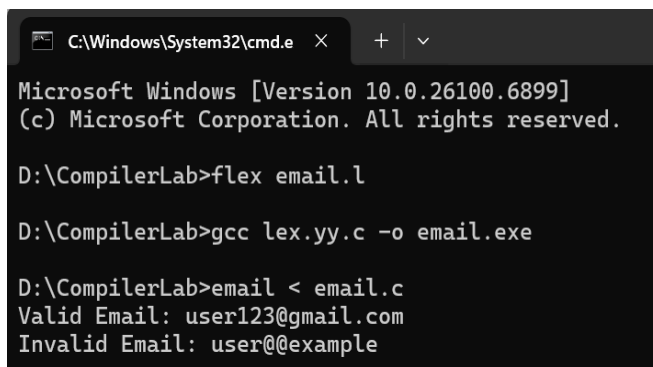
user123@gmail.com

user@@example

Valid Email: user123@gmail.com

Invalid Email: user@@example

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v  
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\CompilerLab>flex email.l  
  
D:\CompilerLab>gcc lex.yy.c -o email.exe  
  
D:\CompilerLab>email < email.c  
Valid Email: user123@gmail.com  
Invalid Email: user@@example
```

25. Write a LEX Program to convert the substring abc to ABC from the given input string.

Code:

```
%{  
#include <stdio.h>  
%}
```

```

%%
abc    { printf("ABC"); }
.\n    { ECHO; }
%%

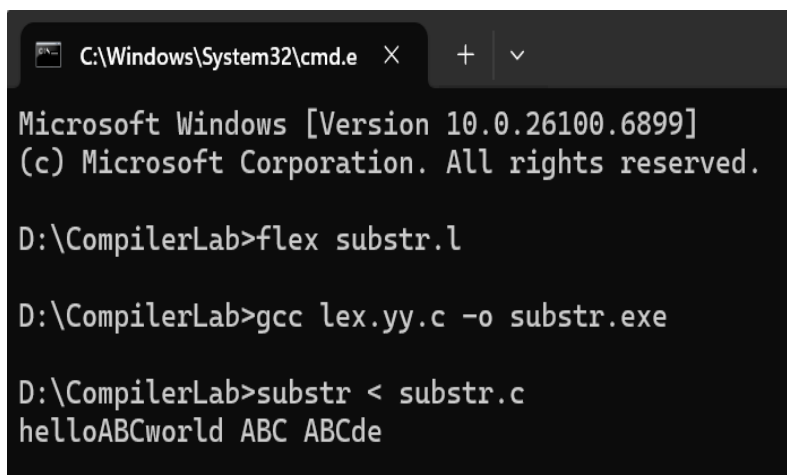
int main(){ yylex(); return 0; }
int yywrap(){ return 1; }

```

Output:

helloabcworld abc abcde
helloABCworld ABC ABCde

Output Screenshot:



```

C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex substr.l

D:\CompilerLab>gcc lex.yy.c -o substr.exe

D:\CompilerLab>substr < substr.c
helloABCworld ABC ABCde

```

26. The Company ABC runs with employees with several departments. The Organization manager had all the mobile numbers of employees. Assume that you are the manager and need to verify the valid mobile numbers because there may be some invalid numbers present. Implement a LEX program to check whether the mobile number is valid or not.

Code:

```

%{
#include <stdio.h>
%}
%%

(91)?[6-9][0-9]{9}    { printf("Valid Mobile Number: %s\n", yytext); }
[0-9]+                { printf("Invalid Mobile Number: %s\n", yytext); }
.\n                   ;
%%

```

```
int main(){ yylex(); return 0; }
```

```
int yywrap(){ return 1; }
```

Output:

9876543210

919876543210

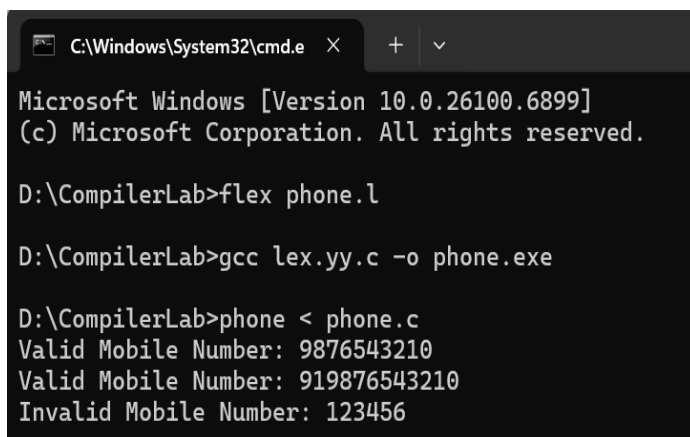
123456

Valid Mobile Number: 9876543210

Valid Mobile Number: 919876543210

Invalid Mobile Number: 123456

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex phone.l

D:\CompilerLab>gcc lex.yy.c -o phone.exe

D:\CompilerLab>phone < phone.c
Valid Mobile Number: 9876543210
Valid Mobile Number: 919876543210
Invalid Mobile Number: 123456
```

27.Implement Lexical Analyzer using LEX or FLEX (Fast Lexical Analyzer). The program should separate the tokens in the given C program and display with appropriate caption.

Code:

```
%{
#include <stdio.h>

%}

IDENT  [a-zA-Z_][a-zA-Z0-9_]*
NUM    [0-9]+
HEADER \<[a-zA-Z0-9._/]+\>

%%

{HEADER}      { printf("Header: %s\n", yytext); }
"int"|"void"|"return"|"printf"|"scanf"|"main" { printf("Keyword: %s\n", yytext); }
{IDENT}       { printf("Identifier: %s\n", yytext); }
```

```

{NUM}      { printf("Number: %s\n", yytext); }
\'([^\\"|\\.)*)\' { printf("String: %s\n", yytext); }
[+\\-*/=,;\\(\\)\\{\\}] { printf("Symbol: %s\n", yytext); }
[ \\t\\n]+  ;
.           ;
%%

int main() { yylex(); return 0; }
int yywrap() { return 1; }

```

Output:

```

#include<stdio.h>

void main(){
    int a,b,c=30;
    printf("hello");
}

```

Header: <stdio.h>

Keyword: void

Keyword: main

Symbol: (

Symbol:)

Symbol: {

Keyword: int

Identifier: a

Symbol: ,

Identifier: b

Symbol: ,

Identifier: c

Symbol: =

Number: 30

Symbol: ;

Keyword: printf

Symbol: (

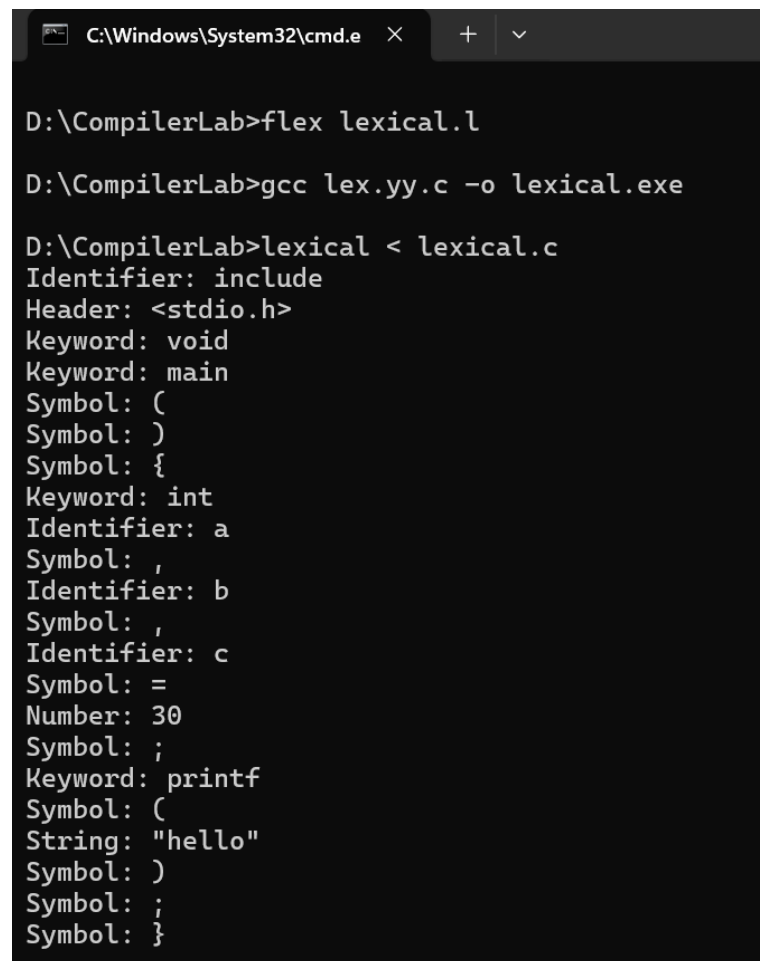
String: "hello"

Symbol:)

Symbol: ;

Symbol: }

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v

D:\CompilerLab>flex lexical.l

D:\CompilerLab>gcc lex.yy.c -o lexical.exe

D:\CompilerLab>lexical < lexical.c
Identifier: include
Header: <stdio.h>
Keyword: void
Keyword: main
Symbol: (
Symbol: )
Symbol: {
Keyword: int
Identifier: a
Symbol: ,
Identifier: b
Symbol: ,
Identifier: c
Symbol: =
Number: 30
Symbol: ;
Keyword: printf
Symbol: (
String: "hello"
Symbol: )
Symbol: ;
Symbol: }
```